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High-gain long-time coherent integration for tones from moving targets via time warping

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ABSTRACT:

The radiated noise from ships is notably characterized by their tonal properties, making tone detection essential for passive sonar systems. The fast Fourier transform plays a crucial role in this process, but Doppler shifts can significantly diminish the integration gain. To address this issue, time warping is applied to modify the time axis of the signal, stabilizing the tonal frequency to a constant value. Traditionally, the warping operator, which determines how to modify the time axis, is designed by estimating motion parameters. In contrast, this study presents a method that eliminates the estimation step by using a set of pre-designed warping operators. The designing process involves preprocessing and clustering the delay vectors corresponding to various motion scenarios. The enhanced gain of the proposed algorithm, in comparison to the existing methods, has been validated through both simulation and experimental data. © 2025 Acoustical Society of America. <https://doi.org/10.1121/10.0036367>

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I. INTRODUCTION

Ship-radiated noise is often regarded as a superposition of strong tones and continuous spectra, where the detection of tones constitutes a critical task for passive sonar systems.¹ With advancements in vibration damping and noise reduction technologies, the source levels of tones for certain acoustic targets have significantly decreased.² In scenarios involving multiple targets and strong interference, the tones are prone to submersion in background noise, thereby limiting the tone detection capability.³ Ideally, tones would exhibit excellent frequency and phase stability,⁴ and could be enhanced through long-time coherent integration. Coherent integration is a method in which signals are accumulated while maintaining their phase alignment. The term “coherent” emphasizes the need to maintain phase relationships within the signal. A key feature of this process is that, while signals can be integrated coherently, noise cannot. As a result, noise components are effectively filtered out, enhancing the signal. This mechanism is particularly valuable for detecting weak signals. However, since the Doppler shift can result in the frequency of the signal being distributed across multiple frequency bins, the gain achieved through coherent integration can be significantly reduced. Thus, it is crucial to explore methods for detecting tones affected by the Doppler effect.

Reference 5 established a peak extraction and multicomponent tracking algorithm for tones based on the low-frequency analysis and recording (LOFAR) through a birth and death process. At low signal-to-noise ratios (SNRs), the high error rate in making frame-by-frame detection decisions

significantly limits the algorithm to identify weak tones. Conversely, the track-before-detect (TBD) algorithms^{6–8} involve accumulating energy to enhance the SNR before making detection decisions, resulting in a notably improved performance under low SNR scenarios. References 9–11 developed tone trackers based on the hidden Markov model (HMM) for extracting tone trajectories from LOFAR spectra. Reference 12 improved tone detection at low SNRs using particle filter-based TBD techniques by real-time estimating noise variance to match the time-varying characteristics of underwater noise. References 13 and 14 enhanced the detection capability for weak tones by combining multi-frame accumulation with TBD. However, TBD techniques are characterized by substantial computational requirements and complex threshold settings. Additionally, these methods typically rely on incoherent integration, leading to a phenomenon known as “SNR threshold,”^{15,16} where the gain from incoherent integration becomes negligible when the initial signal-to-noise (SNR) is too low.

Long-time coherent integration algorithms can mitigate the effects of the “SNR threshold” phenomenon. The Radon Fourier transform (RFT) algorithms,^{17–19} originally developed for the detection of moving radar targets, carry out coherent integration along specific curves in a two-dimensional (2-D) spectrum. A peak associated with the target becomes “focused” when the integration path aligns with the distribution of the signal. Reference 20 introduced this method into the domain of tone detection, developing the long-time coherent integrated line spectrum detection (LTCILSD) algorithm. LTCILSD is an enhancement of the polynomial radon-polynomial Fourier transform (PRPFT) algorithm,²¹ specifically improving the phase compensation

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accuracy during calculation. However, both PRPFT and LTCILSD assume that frequency shifts occur linearly. While this assumption holds in many scenarios, its accuracy declines significantly as the target nears the point of minimum distance along its trajectory. The errors become even more pronounced under conditions of elevated velocities, diminished distances, or extended integration periods. For both PRPFT and LTCILSD, the integral loss resulting from the nonlinear part of frequency shifts cannot be compensated, ultimately limiting the performance.

Time warping^{22–24} offers another way for achieving long-time coherent integration. It essentially performs nonlinear time-domain resampling of the signal, with the warping operator determining the time instances queried during resampling. Reference 25 proposes a cross-power spectral coherent integration algorithm designed specifically for scenarios involving dual hydrophones. This methodology involves estimating the time-scale factor and time delay, subsequently compensating for them accordingly. Reference 26 initially estimates the motion parameters of the target, followed by focusing the energy of the tones through a corresponding time warping. Both approaches described above require estimating the Doppler parameters before Doppler compensation. However, estimating Doppler parameters in conditions of low SNR is challenging, thus limiting the tone detection performance.

This study proposes a matching-based algorithm referred to as the delay-based Doppler compensation (DDC) to address existing challenges in the field. Our research indicates that the creation of a distinct warping operator for each Doppler scenario, which incurs high costs, is unnecessary. Given that numerous scenarios display similar Doppler shifts, employing the same warping operator across these scenarios can yield desirable coherent gains. To implement this strategy, we have developed a set of warping operators that integrate a clustering algorithm. These operators can enhance the coherent integration across a wide range of Doppler shift scenarios. Ultimately, the algorithm eliminates the need for Doppler shift estimation and overcomes the constraints imposed by the linear Doppler shift assumption.

The remainder of this paper is structured as follows: Section II introduces the signal model for tones emitted by targets in uniform linear motion. Section III describes the principles and process flow of the proposed DDC algorithm. Section IV evaluates the coherent integration gains of various algorithms under different Doppler conditions through simulations. Section V validates the effectiveness of the proposed algorithm through sea trial data. Finally, Section VI concludes the research findings.

II. TONE DETECTION AND LONG-TIME COHERENT INTEGRATION

A. The tonal signal model of moving targets

Figure 1 depicts a scenario in which a ship advances in a linear trajectory while maintaining a constant speed v_0 . The sonar is positioned at the origin of the coordinate system, with the x axis aligned with the direction of movement

of the target. The initial angle formed by the line connecting the target and the sonar with respect to the x axis is represented as θ_0 . The initial distance between the target and the sonar is denoted as R_0 . The distance between the target and the sonar at time t is represented as

$$R(t) = \sqrt{R_0^2 + (v_0 t)^2 + 2R_0 v_0 t \cos \theta_0}. \quad (1)$$

The sound emitted by the target is received by the sonar system after a delay given by

$$\tau(t) = R(t)/c, \quad (2)$$

where c represents the speed of sound in water. The signal received by the sonar $s_r(t)$ is a delayed version of the radiated signal of the target, $s(t)$, i.e.,

$$s_r(t) = s(t - \tau(t)). \quad (3)$$

As the target moves, the received signal undergoes either stretching or compression, a phenomenon commonly referred to as the Doppler effect. In scenarios where the target moves radially at a constant velocity, the received signal is uniformly compressed or stretched by a fixed proportion. However, in scenarios involving non-uniform radial movement, the Doppler effect causes more complex distortions in the received signal.

B. Loss of coherent integration

Ideal tones can be represented as the sum of single-frequency components. For simplicity, the discrete single-frequency signal is considered, i.e.,

$$x(n) = A \exp \{j(\omega_1 n + \varphi_1)\}, \quad n = 0, 1, 2, \dots, N-1, \quad (4)$$

where A represents the amplitude, ω_1 denotes the digital angular frequency of the single-frequency signal, φ_1 represents the initial phase, and N implies the length of the signal. The discrete-time Fourier transform (DTFT) of $x(n)$ can be expressed as

$$\begin{aligned} S(\omega) &= \frac{A}{N} \cdot \sum_{n=0}^{N-1} \exp \{j(\omega_1 n + \varphi_1)\} \exp(-j\omega n) \\ &= \frac{A}{N} \cdot \frac{\sin[(\omega - \omega_1)N/2]}{\sin[(\omega - \omega_1)/2]} \\ &\quad \cdot \exp \{j[\varphi_1 + (\omega_1 - \omega)(N-1)/2]\}. \end{aligned} \quad (5)$$

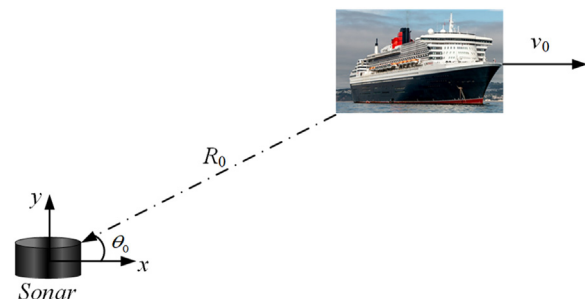


FIG. 1. Target non-radial motion scenario.

Derived from Eq. (5), we establish $|S(\omega_1)| = A$, indicating that DTFT allows for in-phase superposition of the tone. However, in scenarios involving moving targets, the Doppler effect introduces a time-varying phase term $\varphi(n)$. Consequently, the DTFT of the signal is expressed as

$$\begin{aligned} S(\omega) &= \frac{A}{N} \cdot \sum_{n=0}^{N-1} \exp \{j(\omega_1 n + \varphi(n))\} \exp \{-j\omega n\} \\ &= A \cdot S_\varphi(\omega - \omega_1), \end{aligned} \quad (6)$$

where $S_\varphi(\omega)$ represents the DTFT of $\exp \{j\varphi(n)\}$, i.e.,

$$S_\varphi(\omega) = \frac{1}{N} \cdot \sum_{n=0}^{N-1} \exp \{j\varphi(n)\} \exp \{-j\omega n\}. \quad (7)$$

According to Eq. (6), the coherent integration gain depends on $S_\varphi(\omega)$. If $\varphi(n)$ is linear, designated as $\varphi(n) = a_0 + a_1 n$, where a_0 denotes the initial phase and a_1 signifies the rate of phase alteration, it then ensures that $S(\omega_1 + a_1) = A \exp \{ja_0\}$. Therefore, the DTFT continues to achieve an in-phase superposition of the tone. However, in cases where $\varphi(n)$ includes nonlinear terms, the Doppler effect contributes to a loss of integration gain, i.e.,

$$Loss = -\lg(1 - \max \{|S_\varphi(\omega)|\}). \quad (8)$$

C. Brief on PRPFT and LTCILSD

The PRPFT and LTCILSD algorithms assume that the frequencies of the tones are almost linearly shifted²⁰ for targets in uniform linear motion. Thus, the formula of PRPFT is given by

$$\begin{aligned} G(\alpha_1, \alpha_2, \beta_1) &= \frac{1}{N} \sum_{k=0}^{K'-1} X(\alpha_1 + \alpha_2 kD, kD) \\ &\times \exp \left\{ -j \left[\beta_1 kD + \frac{1}{2} \alpha_2 (kD)^2 \right] \right\}, \end{aligned} \quad (9)$$

where α_1 , α_2 , and β_1 represent the search parameters, and K' signifies the number of STFT frames. $X(\alpha_1 + \alpha_2 kD, kD)$ denotes the short-time Fourier transform (STFT) spectrum, which is obtained using a window of length M , and a frame-shift of D . Equation (9) is derived by executing coherent integration over multiple lines within the STFT spectrum. This produces a series of one-dimensional (1-D) spectra, and the one with the highest spectral peak is then selected to identify tones. Due to the limitations of frequency resolution, certain data cannot be directly queried within the STFT spectrum while calculating Eq. (9). The PRPFT algorithm addresses this issue by utilizing nearest-neighbor interpolation, which may introduce computational errors. Conversely, the LTCILSD algorithm seeks to correct these inaccuracies by incorporating fast Fourier transform (FFT) interpolation.

Furthermore, it is worth noting that PRPFT spectrum and LTCILSD spectrum exhibit more severe spectral leakage compared to periodogram. To illustrate this, a specific example is provided. For simplicity, we consider a special case where the Doppler effect is absent. The frequency of the tone is given by $\omega_1 = \omega_0 + \Delta$ where ω_0 denotes the integer multiple of the frequency resolution such that ω_0 is closest to ω_1 , and Δ represents the fractional part. Since the frequency of the tone is constant, the optimal α_2 is 0, as it determines the slope of the integral path. Consequently, the PRPFT spectrum can be expressed as

$$\begin{aligned} S_{\text{PRPFT}}(\alpha_1, \beta_1) &= G(\alpha_1, \alpha_2, \beta_1)|_{\alpha_2=0} = \frac{A \exp \{j\varphi_1\}}{N} \sum_{k=0}^{K'-1} \left(\sum_{n=0}^{D-1} \exp \{j(\omega_0 + \Delta)(n + kD)\} \exp \{-j\alpha_1 n\} \right) \exp \{-j\beta_1 kD\} \\ &= \frac{A \exp \{j\varphi_1\}}{N} \sum_{k=0}^{K'-1} \sum_{n=0}^{D-1} \exp \{j(\omega_0 - \beta_1 + \Delta)kD\} \exp \{j(\omega_0 - \alpha_1 + \Delta)n\}. \end{aligned} \quad (10)$$

To avoid spectral leakage, PRPFT requires both $\Delta = 0$ and $(\omega_0 - \beta_1)D$ to be an integer multiple of 2π simultaneously, whereas the periodogram only requires $\Delta = 0$. Thus, PRPFT is more prone to spectral leakage, potentially reducing spectral peak values and impairing detection efficiency.

D. Brief on time warping

The formula for time warping^{22,23} is expressed as

$$Ux(t) = |\dot{w}(t)|^{1/2} x(w(t)), \quad (11)$$

where $w(t)$ denotes the warping operator, which specifies the time instances queried for resampling. $\dot{w}(t)$ is the first derivative of $w(t)$, which in this study can be fixed at one.

For a discrete signal, the queried time instance is often unavailable. The method of cubic spline interpolation offers a solution. This technique involves fitting discrete data points with a series of cubic polynomials. Figure 2 illustrates a comparison between the interpolated values and the actual values. In this example, the maximum signal frequency is 20 Hz and the sampling rate is 100 Hz. The interpolated values show a close approximation to the actual value and facilitate time warping for discrete signals. It is

essential to note that the sampling rate should be at least five times greater than the signal frequency to ensure interpolation accuracy. If the sampling rate is insufficient, FFT interpolation can be employed to enhance it.

III. THE PROPOSED DDC ALGORITHM

This section details the principles and process flow of the proposed DDC algorithm, including its computational complexity.

A. Design of warping operators

The first step is to build a dataset by listing all the delay vectors for each motion scenario

$$\begin{aligned} \tau(n', i + jN_R + mN_RN_\theta) \\ = \sqrt{R_i^2 + (v_m n' \Delta)^2 + 2R_i v_m (n' \Delta) \cos \theta_j} / c, \\ n' = 0, 1, \dots, N_\tau - 1, \end{aligned} \quad (12)$$

where N_τ denotes the length of sampling time instances n' , Δ denotes the sampling interval, R_i denotes the initial distance, θ_j denotes the initial angle, and v_m denotes the velocity, respectively. They are given by

$$R_i = R_{\min} + i\Delta R, \quad i = 0, 1, \dots, N_R - 1, \quad (13)$$

$$\theta_j = \theta_{\min} + j\Delta\theta, \quad j = 0, 1, \dots, N_\theta - 1, \quad (14)$$

and

$$v_m = v_{\min} + m\Delta v, \quad m = 0, 1, \dots, N_v - 1, \quad (15)$$

where N_R , N_θ , and N_v represent the length of R_i , θ_j , and v_m , respectively.

Moreover, an inspection of Sec. II B reveals that only the nonlinear terms within the delay vectors influence the coherent gain. By eliminating these linear terms, we remove the potential interference for subsequent processing. Figure 3 shows a comparison between delay vectors with and without linear terms. Removing the nonlinear terms reduces the

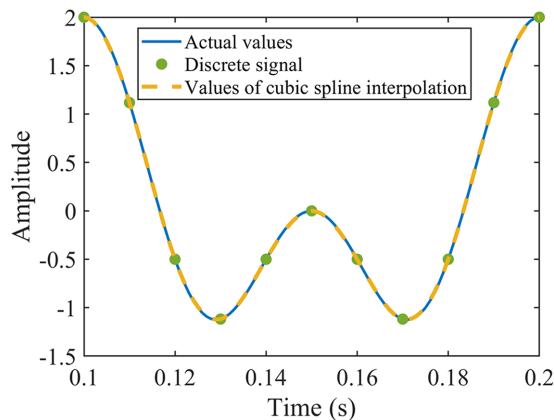


FIG. 2. Comparison of the interpolated values obtained via cubic spline method and the actual values.

Algorithm 1: Scheme of designing the warping operators.

Input: The number of cluster centers P .

Output: The P warping operators $w_p(n)$.

1: Obtain $R_i, i = 0, 1, \dots, N_R - 1; \theta_j, j = 0, 1, \dots, N_\theta - 1; v_m, m = 0, 1, \dots, N_v - 1$ using Eqs. (13)–(15).

2: **For** each $q, q = 0, 1, \dots, N_R N_\theta N_v - 1$

Obtain the delay vector $\tau(n', q)$ using Eq. (12).

Utilize the least squares line fitting method to obtain coefficients of the linear term:

$$a_0, a_1 \leftarrow \text{LSLF}\{\tau(n', q)\}$$

Remove the linear term: $\tau_Q(n', q) \leftarrow \tau(n', q) - a_1 n' - a_0$

End

3: Cluster $\tau_Q(n', q), q = 0, 2, \dots, N_R N_\theta N_v - 1$ using the K -means algorithm to obtain the P cluster centers:

$$[\tau_P(n'), p]_{N_\tau \times P} = K - \text{means}_{N_R N_\theta N_v \rightarrow P}\{\tau_Q(n', q)\}_{N_\tau \times N_R N_\theta N_v}, \text{ where } K - \text{means}\{\cdot\} \text{ denotes the } K\text{-means process.}$$

4: **For** each $p, p = 0, 1, \dots, P - 1$

Adjust the sampling rate to obtain the P warping operators $w_p(n)$:

$$w_p(n) = \text{FFT_interp}_{\Delta \rightarrow 1/F_s}\{\tau_P(n', p)\} + n/F_s, \quad p = 0, 1, \dots, P - 1$$

where $\text{FFT_interp}\{\cdot\}$ denotes FFT interpolation and F_s denotes the sampling of the signal rate.

End

distances between the delay vectors, which, in turn, enhances the accuracy of the clustering center determination.

Linear terms are extracted using the least squares line fitting (LSLF) method. Subsequently, the linear components are removed from the delay vectors. The final outputs are then clustered using K -means²⁷ to derive the warping operators. The methodology for the whole scheme is detailed in Algorithm 1.

Following the designing procedure and the parameters outlined in Table I, the 100 cluster centers are obtained, as shown in Fig. 4(a). Notably, these centers can be precomputed and stored for offline use, thereby reducing the real-time computational demands. To evaluate the degree to which the cluster centers encompass all possible scenarios, we present the statistical distribution of the distances from each delay vector to its nearest cluster center in Fig. 4(b). Approximately 93.2% of delay vectors display a distance

Algorithm 2: Scheme of obtaining the 2-D DDC spectrum.

Input: The precomputed P warping operators $w_p(n)$, the spectrum of the input signal $X(k), k = 0, 1, \dots, K - 1$, the pre-whitening window length O .

Output: The 2-D DDC spectrum

$G_{\text{DDC}}(p, k), k = 0, 1, \dots, K - 1; p = 0, 1, \dots, P - 1$.

1: Calculate the spectral background using a moving average filter of width $O + 1$:

$$X_b(k) = \sqrt{\sum_{i=k-O/2}^{k+O/2} |X(i)|^2}, \quad k = 0, 1, \dots, K - 1$$

2: Pre-whiten the signal: $x_w(n) = \text{IFFT}_{k \rightarrow n}\{X(k)/X_b(k)\}$

3: **For** each $p, p = 0, 1, \dots, P - 1$

Apply time warping and FFTs simultaneously on $x_w(n)$ to obtain the 2-D spectrum:

$$G_{\text{DDC}}(p, k) = \text{FFT}_{n \rightarrow k}\{x_w(w_p(n))\}, \quad p = 0, 1, \dots, P - 1$$

Restore the energy:

$$G_{\text{DDC}}(p, k) = G_{\text{DDC}}(p, k) \cdot X_b(k), \quad k = 0, 1, \dots, K - 1$$

End

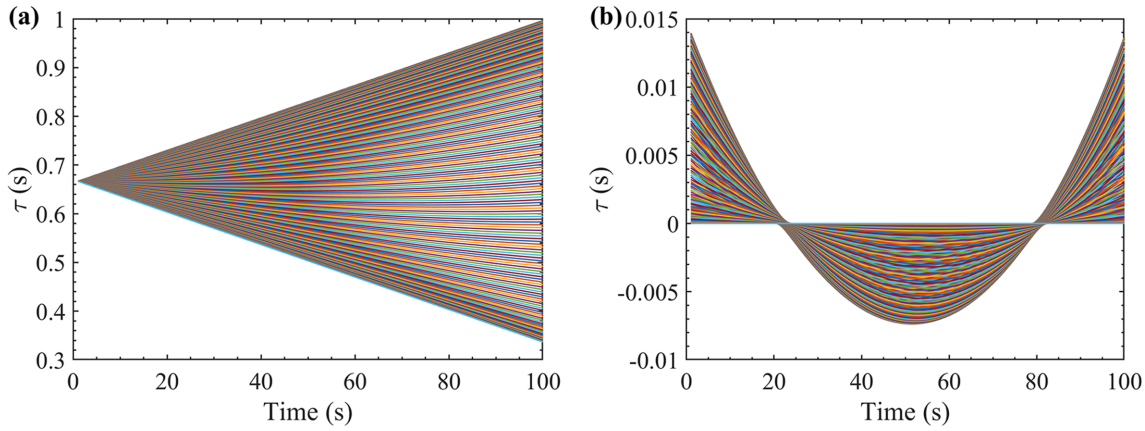


FIG. 3. (a) The delay vectors, (b) the delay vectors that eliminate the linear terms. The delay vectors represent scenarios with an initial distance of 1 km, a velocity of 5 m/s, and initial angles ranging from -180° to 180° , incrementing by 1° .

that falls under 6.9×10^{-5} , which is a negligible level of delay error. For low-frequency tones, a small delay error results in only a minimal phase error when replacing the accurate delay vector with its nearest cluster center.

B. Obtain the 2-D DDC spectrum

This section outlines the process for obtaining the 2-D DDC spectrum, as described in Algorithm 2 and Fig. 3. The first step is to pre-whiten the signal to boost the power of the high-frequency components. Next, the designed warping operators are applied to the signal while simultaneously performing fast Fourier transforms (FFTs) to create a 2-D spectrum. Lastly, the energy is restored.

C. Obtain the 1-D DDC spectrum

This section presents the workflow for obtaining the 1-D DDC spectrum from the 2-D spectrum obtained in Sec. III B. In the case of a single target, the DDC algorithm theoretically requires only one warping operator to focus the tones. Thus, the most straightforward method is using the maximum projection technique, i.e.,

$$k_{\max}, p_{\max} = \arg \max_{k,p} |G_{\text{DDC}}(k,p)|^2, \quad (16)$$

and

$$G_S(k) = |G_{\text{DDC}}(k, p_{\max})|^2. \quad (17)$$

However, when dealing with multiple targets, it becomes necessary to integrate the outputs of multiple warping operators. Reference 20 addresses this through maximum projection along the frequency axis, i.e.,

$$G(k) = \max_p |G_{\text{DDC}}(k,p)|^2. \quad (18)$$

Yet, this method encounters challenges due to the rise in sidelobes as non-optimal warping transformations may yield elevated values adjacent to these tone frequencies. To address this issue, we developed a method which divides the wideband spectrum into several narrow bands based on the local maxima of $G(k)$. Equations (18) and (19) are then applied to each of these narrow bands, treating each narrow band as a single-target instance.

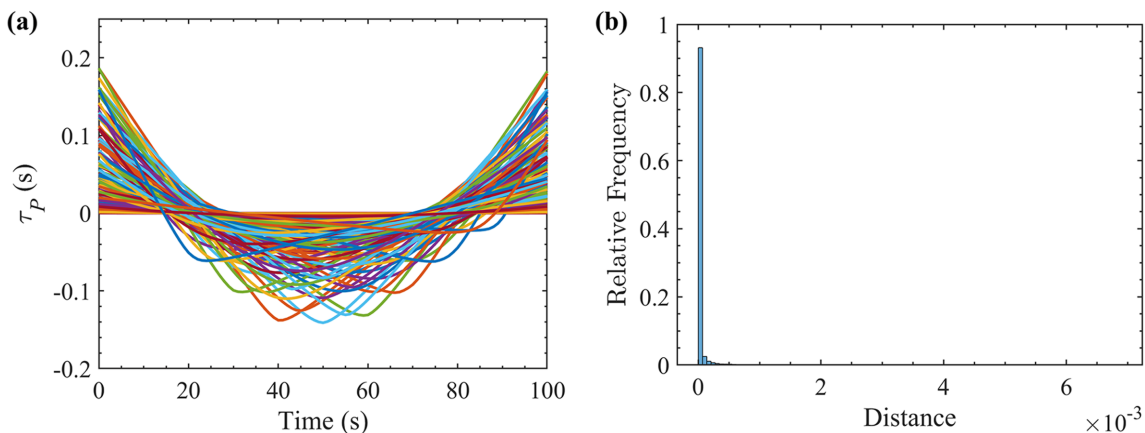


FIG. 4. (a) The 100 cluster centers derived from the K -means algorithm, (b) the statistical distribution of the squared Euclidean distance between each delay vector and its nearest cluster center. A smaller distance indicates a smaller error when replacing the accurate delay vector with its nearest cluster center.

TABLE I. The parameters for designing the warping operators.

Parameter	Δ	P	$R_{\min}$	ΔR	N_R	$\theta_{\min}$	$\Delta\theta$	N_θ	$v_{\min}$	Δv	N_v
Value	1	100	200	100	50	0	1	180	-10	1	21
Unit	s	—	m	m	—	°	°	—	m/s	m/s	—

D. Computational complexity

The computational complexity of the DDC algorithm primarily comes from the cubic spline interpolation and the FFT process. Table II illustrates the computational complexities of the proposed algorithm alongside those of the periodogram, PRPFT, and LTCILSD algorithms. The variables N_{α_1} , N_{α_2} , and N_{β_1} denote the number of search parameters α_1 , α_2 , and β_1 , respectively. The complexity of the DDC algorithm has increased when compared to the PRPFT and LTCILSD algorithms, but it remains within an acceptable range.

IV. SIMULATED EXPERIMENTS ANALYSIS

This section presents a detailed assessment of the proposed algorithm's performance under simulated conditions, compared with the periodogram, PRPFT, and LTCILSD algorithms.

A. Gain comparisons for different methods

The parameters for the experiment are shown in Table III. The tonal frequency is the same as in Ref. 20. The motion parameters are tailored to our specific test environment to demonstrate the scenario in which a significant Doppler shift occurs. Figure 5 presents the LOFAR spectra at different motion velocities, along with the 1-D spectra obtained using different algorithms.

When the target is stationary [Fig. 5(b)], the periodogram shows the highest spectral peak with minimal spectral leakage. However, as the target velocity increases to 5 [Fig. 5(d)], 10 [Fig. 5(f)], and subsequently to 15 m/s [Fig. 5(h)], the Doppler shift of the tones becomes increasingly pronounced, resulting in a significant decrease in the peak value. This significantly increases the difficulty of detection. In contrast, when the target velocities are 5, 10, and 15 m/s, the peak values for the DDC algorithm are -0.68, -0.97, and -1.80 dB, respectively. The peak value changes slightly with an increase in velocity, demonstrating the strong adaptability of the DDC to Doppler shifts. In addition, the relative gains achieved by the DDC algorithm against other methods are as follows: 6.97, 11.54, and

12.6 dB compared to the periodogram; 2.83, 2.06, and 3.82 dB compared to PRPFT; and 1.40, 1.70, and 3.76 dB compared to LTCILSD. The findings demonstrate an enhanced resilience of the DDC algorithm to the Doppler effect. In addition, the spectral peaks of PRPFT and LTCILSD are broader than those of the periodogram and DDC. This is because of the spectral leakage phenomenon, as discussed in Sec. II C.

The advantages of the DDC algorithm are attributed to two main factors. First, the PRPFT and LTCILSD algorithms model the Doppler shift as linear. This simplification can lead to a significant loss of coherence integration gain when the actual conditions diverge significantly. Second, the PRPFT and LTCILSD algorithms, operating within the time-frequency domain, are limited by the time resolution. In every frame, it is not possible to counterbalance the Doppler effect. In brief, the DDC algorithm offers more robust performance in scenarios where the Doppler shift exhibits nonlinear characteristics or when detailed Doppler compensation is necessary.

B. Detection capability for the tones

In order to compare the tone detection performance of the four algorithms under different SNRs, 1000 Monte Carlo experiments were conducted. The probability of detection was statistically evaluated at a false alarm rate of 0.1% in Fig. 6. The simulation parameters are consistent with those described in Sec. IV A. To facilitate a fair comparison, it is essential to align the frequency resolution across different methods prior to detection. This is achieved by taking the maximum value of every μ adjacent frequency points from the high-resolution spectrum, where μ denotes the resolution ratio.

For a static target, the periodogram exhibits the strongest capability in tone detection. The other three methods underperform because they involve deriving the 1-D spectrum for tone detection from a 2-D spectrum, which amplifies the noise components slightly. Consequently, the constant false alarm rate (CFAR) threshold is risen. Moreover, spectral leakage causes the PRPFT and LTCILSD algorithms to experience a reduction in the peak

TABLE II. Complexity of different algorithms.

Algorithm	Multiplications	Additions
Periodogram	$0.5N \log_2 N$	$N \log_2 N$
PRPFT	$0.5N \log_2 M + N_{\alpha_1} N_{\alpha_2} K (0.5 \log_2 K + 1)$	$N \log_2 M + N_{\alpha_1} N_{\alpha_2} K \log_2 K$
LTCILSD	$N \log_2 M + N_{\beta_1} N_{\alpha_2} N + 0.5 N_{\alpha_1} N_{\alpha_2} \log_2 N_{\alpha_1}$	$2N \log_2 M + N_{\alpha_1} N_{\alpha_2} \log_2 N_{\alpha_1}$
DDC	$7N + 6PN + 0.5PN \log_2 N$	$6N + 6PN + PN \log_2 N$

TABLE III. Parameters for the simulation experiment. f_0 denotes the frequency of the tone and T denotes the integration duration.

Parameter	F_s	f_0	T	R_0	θ_0	M	D	O
Value	1000	100.02	100	1855	85	20	20	500
Unit	Hz	Hz	s	m	°	s	s	—

values. As a result, these two algorithms exhibit reduced detection capability compared to the proposed DDC algorithm.

For a moving target with velocities of 5, 10, and 15 m/s, the DDC algorithm demonstrates the highest tone detection performance, followed by the LTCILSD algorithm. To achieve an 80% detection probability, the DDC algorithm requires minimum SNR values of -4.3 , -4.2 , and -3.7 dB, respectively. These values are 1, 1.25, and 3.5 dB lower than those required by the LTCILSD algorithm. Additionally, as the target velocity increases from 5 to 15 m/s, the minimum SNR required by the DDC algorithm increases by only about 0.6 dB, highlighting its strong resilience to Doppler shifts.

C. Gain comparison under different motion scenarios

The loss of coherent integration, termed as $Loss$ and calculated by Eq. (8), can be utilized to evaluate the performance of the coherent integration algorithm. Figure 7 illustrates the coherent integration losses for different motion scenarios. R_0 and θ_0 were discretized into 0, 50, 100, ..., 10 000 and 0, 1, 2, ..., 180, respectively.

To assess the gains achieved through different methods, we employ $L_{\max}$, which denotes the maximum loss in coherent integration across the defined parameter space (R_0, θ_0). Moreover, we examine the area where the coherent integration loss is less than -3 dB, i.e.,

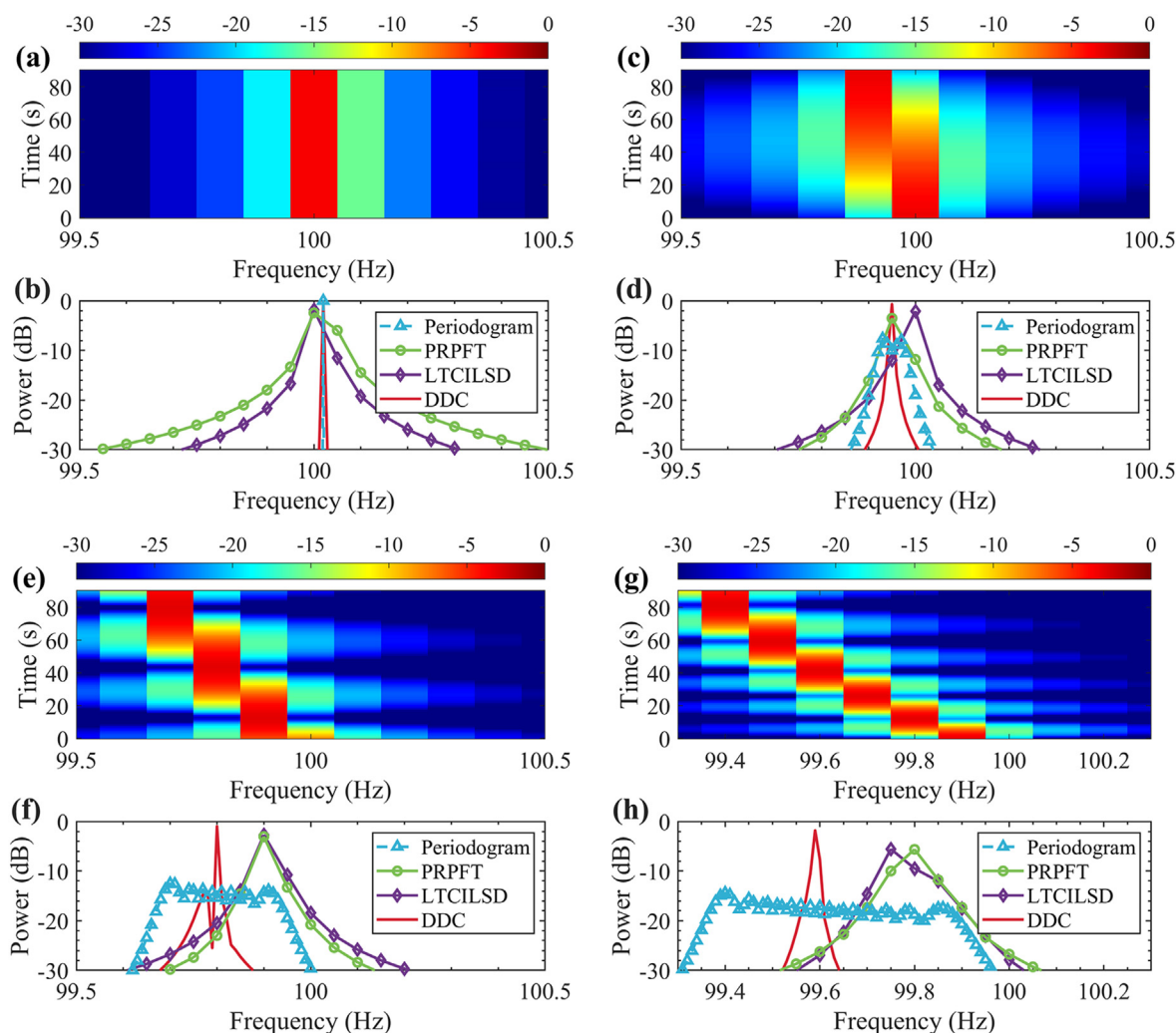


FIG. 5. LOFAR spectra and 1-D spectra obtained with different algorithms. (a), (c), (e), and (g) represent LOFAR spectra at velocities of 0, 5, 10, and 15 m/s, respectively. (b), (d), (f), and (h) represent 1-D spectra at velocities of 0, 5, 10, and 15 m/s, respectively.

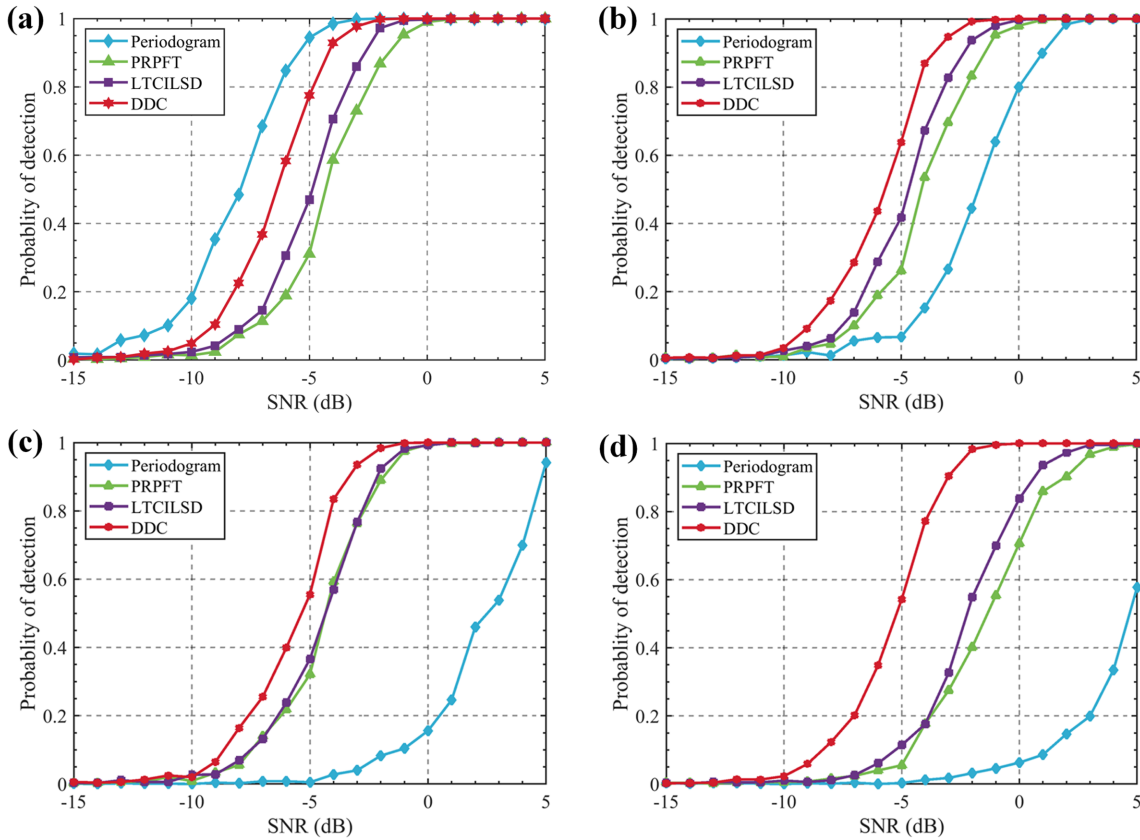


FIG. 6. Detection probabilities across various SNRs for the 100 Hz tone. The target velocities are (a) 0, (b) 5, (c) 10, and (d) 15 m/s, respectively. The false alarm rate is 0.1%.

$$S_{-3\text{dB}} = \frac{1}{N_R N_\theta} \sum_{m=1}^{N_\theta} \sum_{n=1}^{N_R} I(\mathbf{L}_{m,n}), \quad (19)$$

where

$$I(x) = \begin{cases} 1, & x \geq -3, \\ 0, & x < -3. \end{cases} \quad (20)$$

A lower value of $L_{\max}$ combined with a higher value of $S_{-3\text{dB}}$ suggests that the algorithm exhibits increased resilience to Doppler shifts. Table IV provides a comparative analysis of $L_{\max}$ and $S_{-3\text{dB}}$ across different methods.

The proposed DDC algorithm demonstrates superior performance when compared to the existing methods. Specifically, for targets moving at velocities of 5, 10, and 15 m/s, the DDC algorithm attained $L_{\max}$ values of -6.45 , -7.64 , and -8.54 dB, respectively. These outcomes not only surpass those obtained through the periodogram method by margins of 7.66, 9.04, and 9.82 dB but also exceed the results obtained using the LTCILSD algorithm by 3.18, 5.19, and 6.72 dB, respectively. Upon comparison, it is evident that the LTCILSD algorithm shows limited improvements in scenarios involving large Doppler shifts. In contrast, the DDC algorithm effectively mitigates losses in such scenarios, ensuring a robust coherent integration gain.

Additionally, the values of $S_{-3\text{dB}}$ are 98.1%, 92.1%, and 82.9% for the DDC algorithm, respectively. These values

surpass those observed with the periodogram by 34.8%, 65.1%, and 64.0%, and are higher than those achieved by the LTCILSD by 3.3%, 4.0%, and 5.6%, respectively. The LTCILSD and DDC algorithms are both specifically designed for the analysis of moving targets, thereby significantly improving the coherent integration gain for moving targets. Compared to the LTCILSD algorithm, the DDC algorithm may not outperform in every specific case, but it offers clear advantages on average, especially in scenarios where Doppler shifts exhibit nonlinear characteristics.

V. SEA TRIAL ANALYSIS

This section presents an evaluation of the proposed DDC algorithm's performance using sea trial data.

A. Experiment description

In this study, data collected by the first array element of the vertical line array (VLA) in the SWellEx-96 Event S59²⁸ were selected for analysis. As illustrated in Fig. 8, the source ship (R/V Sproul) moved northward at a constant speed of approximately 2.5 m/s. The distance between the sound source ship and the sonar system progressively increased from 1.2 to 4.6 km. The ship towed two different sources: a "deep water" source (J-15) and a "shallow water" source (J-13), which emitted a series of tones across a

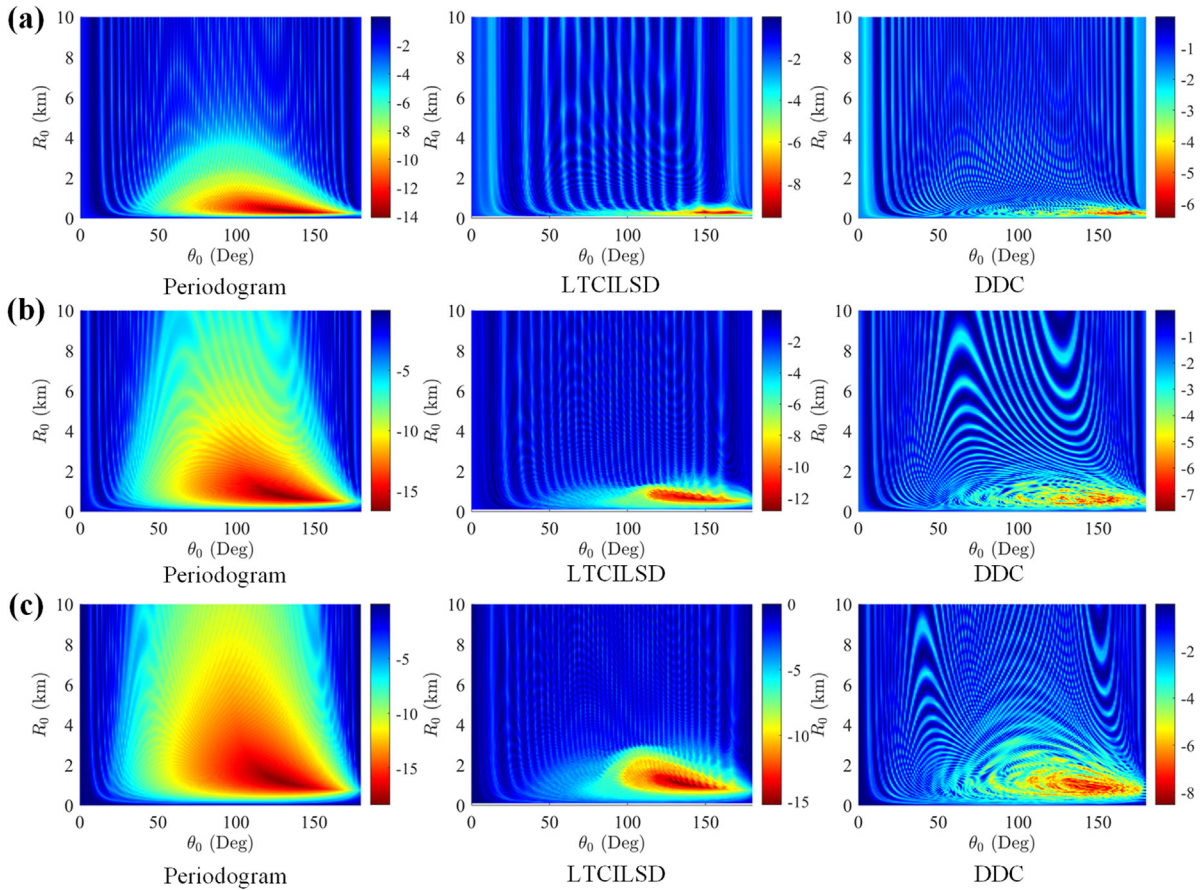


FIG. 7. Coherent integration loss (dB) for different methods across different initial distances, R_0 , and initial angles, θ_0 . The target velocities are (a) 5, (b) 10, and (c) 15 m/s, respectively. The loss is calculated for a 100 Hz tone over a duration of 100 s.

TABLE IV. Comparison of resilience to the Doppler effect across different methods.

	$L_{\max}$ (dB)			$S_{-3\text{dB}}$ (%)		
	5	10	15	5	10	15
Periodogram	-14.11	-16.68	-18.36	63.3	27.0	18.9
LTCILSD	-9.63	-12.83	-15.26	94.8	88.1	77.3
DDC	-6.45	-7.64	-8.54	98.1	92.1	82.9

TABLE V. Relative gains (in dB) achieved by the proposed DDC algorithm compared to other algorithms over a 100 s integration period, with the frequency range from 125 to 146 Hz.

Frequency (Hz)	127	130	135	140	145
$G_{\text{DDC-Periodogram}}$	3.5	4.3	0.0	3.7	3.9
$G_{\text{DDC-PRPFT}}$	2.0	2.7	-0.2	0.2	1.7
$G_{\text{DDC-LTCILSD}}$	0.6	2.2	0.4	0.0	-0.1

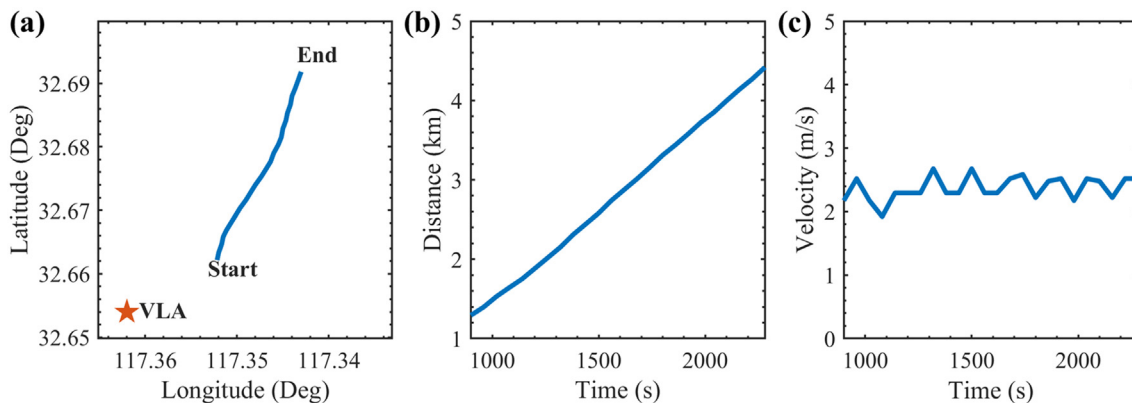


FIG. 8. Motion parameters of the sound source ship: (a) trajectory, (b) distance to the sonar system, (c) velocity.

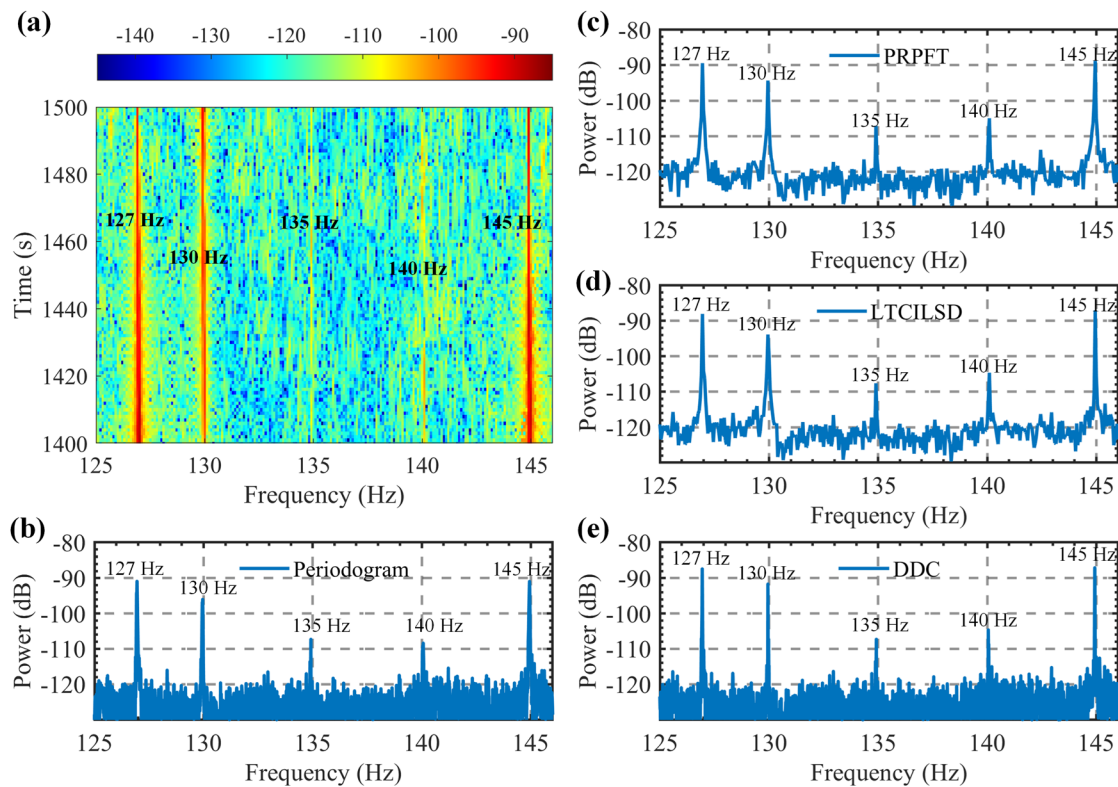


FIG. 9. Processing results of sea trial data within the 125 to 146 Hz frequency band. (a) LOFAR spectrum; 1-D spectrum attained via (b) periodogram, (c) PRPFT, (d) LTCILSD, (e) DDC. The integration time is 100 s.

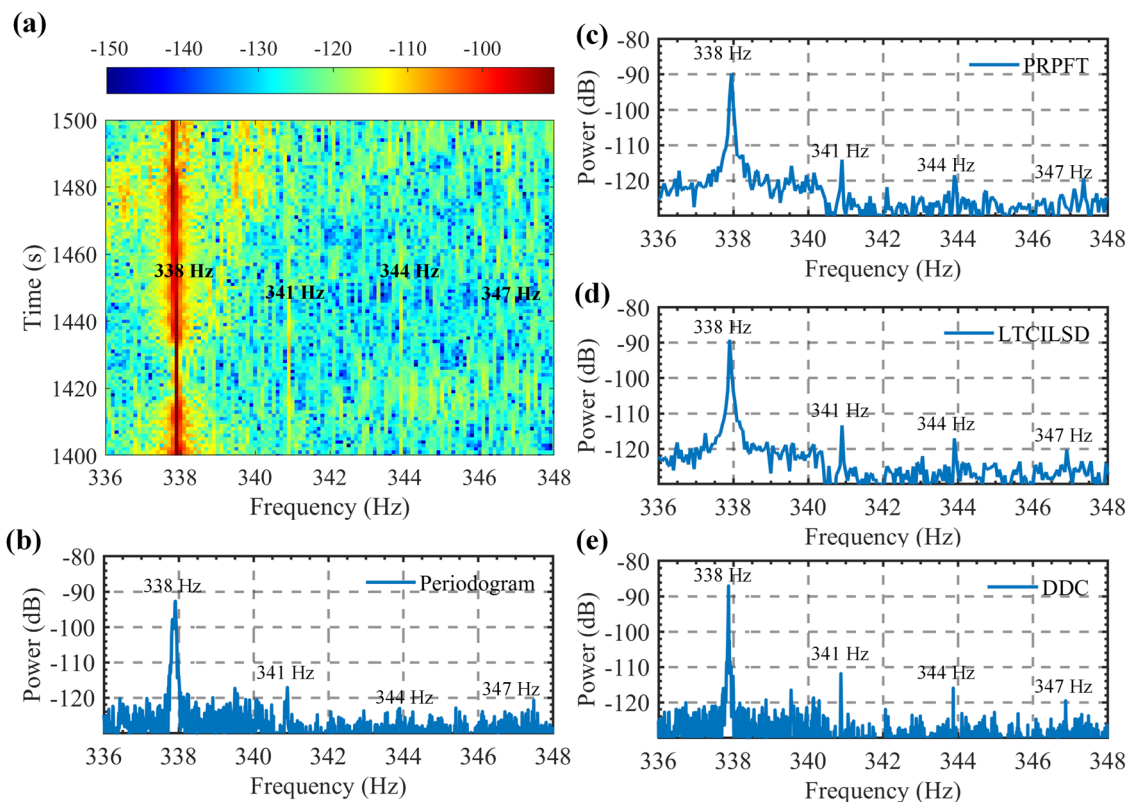


FIG. 10. Processing results of sea trial data within the 336 to 348 Hz frequency band. (a) LOFAR spectrum; 1-D spectrum attained via (b) periodogram, (c) PRPFT, (d) LTCILSD, (e) DDC. The integration time is 100 s.

TABLE VI. The relative gains (in dB) achieved by the proposed DDC algorithm compared to other algorithms over a 100 s integration period, with the frequency range from 336 to 348 Hz.

Frequency (Hz)	338	341	344	347
$G_{\text{DDC-Periodogram}}$	5.6	5.2	7.1	5
$G_{\text{DDC-PRPFT}}$	2.7	2.2	2.6	4.4
$G_{\text{DDC-LTCILSD}}$	2.1	1.4	1.1	0.7

frequency range of 49 to 400 Hz. These tones were recorded at a sampling rate of 1.5 kHz.

B. Effect of 100 s coherent integration

The experimental data were processed using the periodogram, PRPFT, and LTCILSD algorithms to compare their efficacy. Figure 9 shows the LOFAR spectra within the 125 to 146 Hz frequency band, as well as the power spectra obtained through 100 s coherent integration. The method described in Sec. III C was employed to obtain the 1-D spectrum.

It is observed that within the frequency range of 125 to 146 Hz, five prominent tones are present at frequencies of 127, 130, 135, 140, and 145 Hz. The tones at 127, 130, and 145 Hz are from the ship towed sources, while the tones at 135 and 140 Hz are from unknown sources. The noise spectral envelopes produced by the algorithms are almost identical. This similarity allows us to overlook the impact of algorithms on the noise and concentrate on their efficacy in enhancing coherence of tones in subsequent analyses. Table V illustrates the relative gains achieved by the proposed DDC algorithm. Specifically, $G_{\text{DDC-Periodogram}}$, $G_{\text{DDC-PRPFT}}$, and $G_{\text{DDC-LTCILSD}}$ signify the relative gain (measured in dB) that the DDC algorithm achieves over the periodogram, PRPFT, and LTCILSD algorithm, respectively.

The DDC algorithm demonstrates enhancements compared to the periodogram. The relative gains for the five

tones are 3.5, 4.3, 0.0, 3.7, and 3.9 dB, respectively, resulting in an average improvement of 3.08 dB. The lack of improvement for the tone at 135 Hz can be primarily attributed to its minimal Doppler shift. In comparison to PRPFT, the enhancements provided by the DDC algorithm are 2.0, 2.7, -0.2, 0.2, and 1.7 dB, resulting in an average increase of 1.28 dB. Compared with LTCILSD, the DDC algorithm presented increases of 0.6, 2.2, 0.4, 0.0, and -0.1 dB, with an overall average gain of 0.62 dB.

Subsequently, the efficiency of the algorithms in analyzing higher-frequency tones was evaluated. Figure 10 illustrates the processing results within the 336 to 348 Hz frequency band, featuring four tones at frequencies of 338, 341, 344, and 347 Hz.

In the periodogram, the tones at 338 and 341 Hz are prominently observable, while the tones at 344 and 347 Hz are no longer detectable. However, all four tones are discernible in LTCILSD spectrum and DDC spectrum. Table VI compares the relative gains of the DDC algorithm against the other methods. By averaging the results, it is found that the relative gain provided by the DDC algorithm compared to the periodogram, PRPFT, and LTCILSD, is 5.7, 3.0, and 1.3 dB, respectively.

C. Effect of 400 s coherent integration

In this section, a 400 s coherent integration was performed to analyze the impact of the algorithm on the tones within the frequency range of 125 to 146 Hz (Fig. 11).

Table VII presents the relative gains realized by the proposed DDC algorithm against those achieved by the periodogram, PRPFT, and LTCILSD algorithms, with the DDC algorithm showing average values of 5.12, 2.24, and 2.08 dB, respectively. This comparison highlights the outstanding performance of the proposed algorithm, particularly when applied over very long coherent integration periods. As the duration of coherent integration increases, the errors arising from the

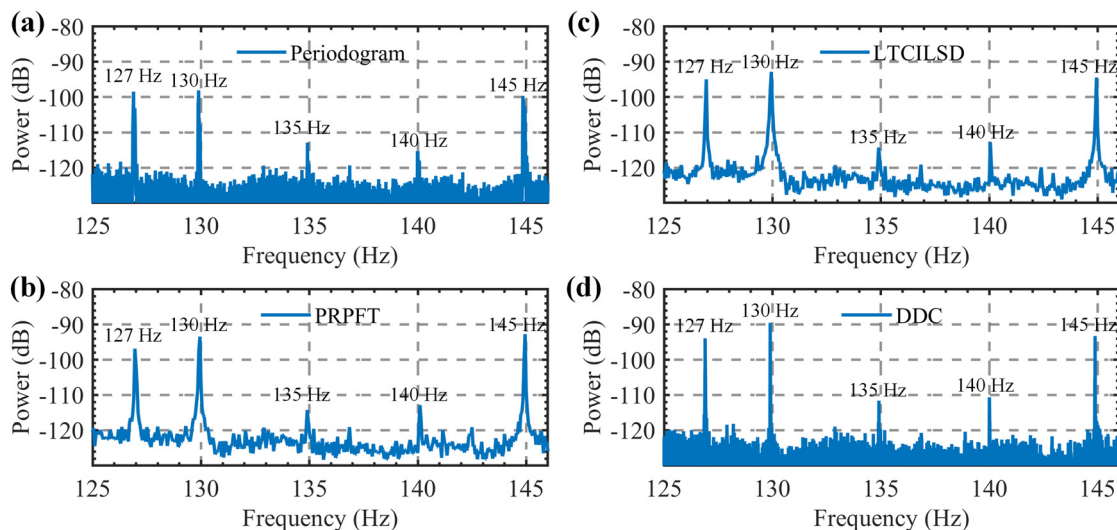


FIG. 11. 1-D spectrum within the frequency range of 125 to 146 Hz attained via (a) periodogram, (b) PRPFT, (c) LTCILSD, and (d) DDC. The integration time is 400 s.

TABLE VII. The relative gain (in dB) achieved by the proposed DDC algorithm compared to other algorithms over a 400 s integration period, with the frequency range from 125 to 146 Hz.

Frequency (Hz)	127	130	135	140	145
$G_{\text{DDC-Periodogram}}$	4.6	8.6	1.3	4.6	6.5
$G_{\text{DDC-PRPFT}}$	2.9	3.9	2.7	2.2	-0.5
$G_{\text{DDC-LTCILSD}}$	1.1	3.3	2.7	2	1.3

assumptions of linear Doppler shift become more significant. In contrast, the proposed DDC algorithm does not rely on this specific assumption and demonstrates enhanced robustness against nonlinear Doppler shifts.

D. Extension to envelope spectrum analysis

This section extends the DDC to envelope spectrum²⁹ analysis. Spectral peaks become evident in the envelope spectrum when the envelope of the signal displays periodicity. The formula for envelope spectrum is given by

$$E_x(k) = \sum_{n=0}^{N-1} |x_b(n)|^2 \exp \{-j2\pi kn/N\}, \quad (21)$$

where $x_b(n)$ denotes the signal that passes through a band-pass filter. The extension of DDC to envelope spectrum analysis can be easily realized by performing analysis on the warped signal $x_{\text{opt}}(n) = x(w_{\text{opt}}(n))$, where $w_{\text{opt}}(n)$ denotes the optimal warping operator to produce the highest spectral peak. Figure 12 depicts the comparison of the envelope spectrum obtained from the original signal and the warped signal. The data used is as detailed in Sec. VB.

It is evident that time warping significantly enhances the harmonics featuring a fundamental frequency of 16.16 Hz. It demonstrates that DDC can be easily extended to envelope spectrum analysis. In contrast, as PRPFT and

LTCILSD fail to produce an improved time-domain signal, their extension to envelope spectrum analysis poses significant challenges.

VI. CONCLUSION

In this paper, we have presented a systematic analysis on the factors that affect coherent integration gain of tones emitted from uniformly linearly moving targets. Subsequently, a long-time coherent integration algorithm via time warping is proposed for detecting tones, named DDC. The K -means algorithm has been used in the design of the warping operators for reducing the computational load, while ensuring the resilience of the algorithm to the Doppler effect. DDC exhibits superior performance compared to existing methods, particularly in scenarios where the Doppler shift displays nonlinear characteristics. Both simulations and experiments have demonstrated the efficacy of DDC in achieving high-gain coherent integration. The proposed algorithm is limited to uniform linear motion. Future work could extend it to include more complex motion models. Additionally, future research may combine PRPFT and time warping to further improve Doppler resilience.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are openly available in the UC San Diego Library Digital Collections at <https://doi.org/10.6075/J0MW2F21>, SWellEx-96 Experiment Acoustic Data.

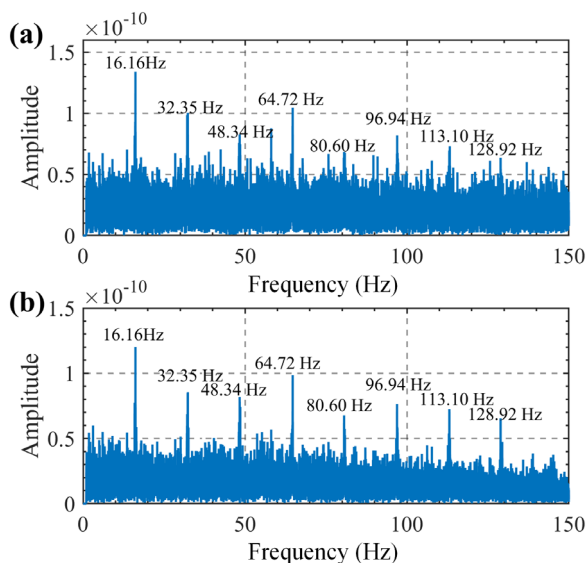


FIG. 12. The envelope spectra based on (a) the original signal, (b) the warped signal. The frequency range of the bandpass filter is 500 to 1500 Hz.

- ¹Y. Cheng, Z. Li, and J. Qiu, "Line spectrum characteristics of ship radiated noise," in *Underwater Acoustic Target Recognition*, 1st ed. (Science Press, Beijing, 2018), pp. 66–67.
- ²T. A. Smith and J. Rigby, "Underwater radiated noise from marine vessels: A review of noise reduction methods and technology," *Ocean Eng.* **266**, 112863 (2022).
- ³G. Liang, Y. Chen, J. Wang, Y. Li, and L. Qiu, "Enhanced noise resilience in passive tone detection via broad-receptive field complex-valued convolutional neural networks," *J. Acous. Soc. Am.* **155**, 3968–3982 (2024).
- ⁴S. Burenkov, A. Gavrilov, A. Uporin, and A. Furduev, "Heard Island feasibility test: Long-range sound transmission from Heard Island to Krylov underwater mountain," *J. Acous. Soc. Am.* **96**, 2458–2463 (1994).
- ⁵Z. Li and N. Martin, "A time-frequency based method for the detection and tracking of multiple non-linearly modulated components with births and deaths," *IEEE Trans. Signal Process.* **64**, 1132–1146 (2016).
- ⁶L. A. Johnston and V. Krishnamurthy, "Performance analysis of a dynamic programming track before detect algorithm," *IEEE Trans. Aerosp. Electron. Syst.* **38**, 228–242 (2002).
- ⁷Y. Boers and J. Driessen, "Multitarget particle filter track before detect application," *IEEE Proc. Radar. Sonar Navig.* **151**, 351–357 (2004).

- ⁸M. Kumar and S. Mondal, "Recent developments on target tracking problems: A review," *Ocean Eng.* **236**, 109558 (2021).
- ⁹X. Luo, L. Li, and Z. Shen, "Line spectrum trajectory extraction method of underwater acoustic signal based on dynamic parameter HMM," *J. Electron. Inf. Technol.* **44**, 1956–1965 (2022).
- ¹⁰X. Luo and Z. Shen, "A space-frequency joint detection and tracking method for line-spectrum components of underwater acoustic signals," *Appl. Acoust.* **172**, 107609 (2021).
- ¹¹X. Xie and R. J. Evans, "Multiple target tracking and multiple frequency line tracking using hidden markov models," *IEEE Trans. Signal Process.* **39**, 2659–2676 (1991).
- ¹²D. Zhang, L. Gao, D. Sun, and M. Teng, "Soft-decision detection of weak tonals for passive sonar using track-before-detect method," *Appl. Acoust.* **188**, 108549 (2022).
- ¹³L. Zhang, S. Piao, J. Guo, and X. Wang, "Multi-frame coherent track-before-detect method for weak tones in passive sonar," *Signal Process.* **220**, 109437 (2024).
- ¹⁴L. Zhang, S. Piao, J. Guo, and X. Wang, "Multi-frame non-coherent track-before-detect method for weak tones in passive sonar," *J. Phys. Conf. Ser.* **2718**, 012086 (2024).
- ¹⁵R. Tandra and A. Sahai, "Fundamental limits on detection in low SNR under noise uncertainty," in *2005 International Conference on Wireless Networks, Communications and Mobile Computing*, HI (IEEE, New York, 2005), pp. 464–469.
- ¹⁶R. Tandra and A. Sahai, "SNR walls for signal detection," *IEEE J. Sel. Top. Signal Process.* **2**, 4–17 (2008).
- ¹⁷J. Xu, J. Yu, Y.-N. Peng, and X.-G. Xia, "Radon-Fourier transform for radar target detection, I: Generalized Doppler filter bank," *IEEE Trans. Aerosp. Electron. Syst.* **47**(2), 1186–1202 (2011).
- ¹⁸J. Yu, J. Xu, Y. N. Peng, and X. G. Xia, "Radon-Fourier transform for radar target detection (III): Optimality and fast implementations," *IEEE Trans. Aerosp. Electron. Syst.* **48**, 991–1004 (2012).
- ¹⁹J. Xu, J. Yu, Y. Peng, and X. Xia, "Focus-before-detect for radar target detection," in *Proceedings of 2011 IEEE CIE International Conference on Radar*, Chengdu, China (IEEE, New York, 2011), pp. 430–433.
- ²⁰L. Zhang, S. Piao, and J. Guo, "Passive tone detection for moving targets based on long-time coherent integration," *IEEE J. Ocean. Eng.* **48**(3), 820–836 (2023).
- ²¹W. Wu, G. Wang, and J. Sun, "Polynomial radon-polynomial Fourier transform for near space hypersonic maneuvering target detection," *IEEE Trans. Aerosp. Electron. Syst.* **54**, 1306–1322 (2018).
- ²²D.-Y. Gao, D.-Z. Gao, C. Jing, W. Liang, and S. Wen-Hua, "Doppler-warping transform and its application to estimating acoustic target velocity," *Acta Phys. Sin.* **70**, 124302 (2021).
- ²³C. Ioana, A. Quinquis, and Y. Stephan, "Feature extraction from underwater signals using time-frequency warping operators," *IEEE J. Ocean. Eng.* **31**, 628–645 (2006).
- ²⁴J. Bonnel, A. Thode, D. Wright, and R. Chapman, "Nonlinear time-warping made simple: A step-by-step tutorial on underwater acoustic modal separation with a single hydrophone," *J. Acoust. Soc. Am.* **147**, 1897–1926 (2020).
- ²⁵W. Guo, S. Piao, J. Guo, Y. Lei, and K. Iqbal, "Passive detection of ship-radiated acoustic signal using coherent integration of cross-power spectrum with Doppler and time delay compensations," *Sensors* **20**, 1767–1787 (2020).
- ²⁶J. Mao, Z. Peng, L. Zhang, G. Wang, and K. Liu, "Line spectrum detection of airborne source from underwater using Doppler-warping transform," *Acta Acust.* **47**, 810–819 (2022).
- ²⁷A. M. Ikotun, A. E. Ezugwu, L. Abualigah, B. Abuhaija, and J. Heming, "K-means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data," *Inf. Sci.* **622**, 178–210 (2023).
- ²⁸N. O. Booth, W. S. Hodgkiss, and D. E. Ensberg, "SWellEx-96 Experiment Acoustic Data," UC San Diego Library Digital Collections. Dataset (2015).
- ²⁹P. Clark, I. Kirsteins, and L. Atlas, "Multiband analysis for colored amplitude-modulated ship noise," in *2010 IEEE International Conference on Acoustics, Speech and Signal Processing*, Dallas, TX (IEEE, New York, 2010), pp. 3970–3973.